# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02**

**COURSE OUTCOME COVERED**
***CO2:*** *Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analysing images.. (BL3)*

*Assessment Tool Weightage: 30%*

**QUESTIONS: 3 Total Marks:30** Annexure A

# Scenario-Based

***Code of Conduct:***

*I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

RUBRICS FOR CALCULATION

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| **Criteria** | **Weight age (%)** | **Level 4 (Exemplary)** | **Level 3 (Proficient)** | **Level 2 (Developing)** | **Level 1 (Beginning)** | **Marks** |
| Understa nding of Scenario | 20% | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps | Partial understanding; misses key aspects | Misinterprets or fails to understand the scenario | |
| Identifi ca tion of Key Issues | 20% | Accurately identifies all relevant issues and factors involved | Identifies most key issues with minor omissions | Identifies limited issues; some irrelevant points | Unable to identify key issues | |
| Applic ati on of Concepts | 25% | Applies appropriate concepts effectively to address the scenario | Applies concepts with minor errors | Limited or partially correct application | Unable to apply relevant concepts | |

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|---|---|---|---|---|---|---|
| Decision Making | 20% | Makes well-reasoned, logical decisions with strong rationale | Makes reasonable decisions with some justification | Makes weak decisions with limited justification | No clear decisions made or rationale provided | |
| Clarity of Response | 15% | Response is clear, structured, and logically presented | Generally clear with minor issues | Somewhat unclear or poorly structured | Disorganized and difficult to understand | |

Annexure A

# Scenario-Based

### 1. Crack Detection on Concrete Surfaces Using Drone Images

**Scenario:** A civil inspection team is developing a computer vision system to detect thin cracks on concrete bridges and building surfaces. Images are captured using a drone under outdoor conditions. Due to changing sunlight, shadows, and camera vibration, the raw images contain uneven illumination, low contrast, and Gaussian noise.

**Questions:**

**A.** Propose a suitable image preprocessing pipeline to prepare the drone images for crack detection. Your answer should include methods for illumination correction, noise reduction, contrast enhancement, and image normalization. Justify the need for each step.

**B.** Select an appropriate edge detection technique for identifying thin surface cracks. Compare it with at least one other edge detection method and explain why your selected technique is more suitable for detecting fine cracks.

**C.** After edge detection, some crack segments appear broken or disconnected. Suggest a suitable post-processing technique to connect the crack regions and improve the final detection output.

### 2. Human Motion Tracking in a Security Camera Video

**Scenario:** A security camera is installed in a hallway to monitor people moving in different directions. The system must estimate the motion of each person across video frames for tracking and activity analysis. The hallway contains plain walls, smooth floors, and occasional shadows.

**Questions:**

**A.** Explain how the Lucas-Kanade optical flow method can be used to estimate the motion of people between consecutive video frames.

**B.** Discuss how the aperture effect can create errors in motion estimation, especially on uniform regions such as plain walls, floors, and clothing surfaces.

**C.** If a person suddenly changes speed or direction, describe how the optical flow vectors will be affected. Suggest a method to improve tracking accuracy in such situations.

### 3. Vehicle Tracking at a Traffic Junction Using Video Frames

**Scenario:** A smart traffic monitoring system uses CCTV footage to track vehicles at a busy road junction. Vehicles move at different speeds, some stop suddenly at signals, and others turn left or right. The video also contains shadows, reflections, and partial occlusion when vehicles overlap.
**Questions:**

**A.** Explain how optical flow can be used to estimate the direction and speed of moving vehicles in the video sequence.

**B.** Identify two challenges that may affect accurate motion estimation in this scenario, such as occlusion, shadows, sudden braking, or camera vibration. Explain how each challenge affects the optical flow result.

**C.** Suggest a suitable improvement method to make vehicle tracking more reliable when vehicles overlap or change direction suddenly.

## 1. Crack Detection on Concrete Surfaces Using Drone Images

### A. Image Preprocessing Pipeline

A suitable preprocessing pipeline is:

**Step 1: Image Acquisition**

- Capture high-resolution drone images of the concrete surface.
- **Need:** Provides clear images for accurate crack detection.

**Step 2: Illumination Correction**

- Apply **Histogram Equalization** or **CLAHE (Contrast Limited Adaptive Histogram Equalization)**.
- **Need:** Corrects uneven lighting caused by sunlight and shadows, making cracks more visible.

**Step 3: Noise Reduction**

- Apply a **Gaussian Filter**.

- **Need:** Removes Gaussian noise caused by camera vibration and sensors while preserving the overall crack structure.

**Step 4: Contrast Enhancement**

- Apply **Contrast Stretching** or **CLAHE**.
- **Need:** Improves the visibility of thin cracks against the concrete background.

**Step 5: Image Normalization**

- Normalize pixel intensity values to a fixed range (e.g., 0–255 or 0–1).
- **Need:** Ensures consistent brightness and contrast across all images, improving detection accuracy.

**Step 6: Edge Detection**

- Detect crack boundaries using an edge detector such as **Canny**.

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**Summary of the Pipeline**

1. Image Acquisition
2. Illumination Correction (Histogram Equalization/CLAHE)
3. Gaussian Noise Reduction
4. Contrast Enhancement (CLAHE/Contrast Stretching)
5. Image Normalization
6. Edge Detection

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**B. Suitable Edge Detection Technique**

**Selected Technique: Canny Edge Detector**

**Reasons:**

- Detects very thin cracks accurately.
- Produces thin, well-localized edges.
- Uses Gaussian smoothing to reduce noise.
- Uses non-maximum suppression to avoid thick edges.
- Uses double thresholding to distinguish strong and weak edges.

**Comparison with Sobel Edge Detector**

| Feature | Canny | Sobel |
|---|---|---|
| Noise Handling | Excellent | Moderate |
| Edge Thickness | Thin | Thick |
| Edge Localization | High | Moderate |
| Detects Fine Cracks | Excellent | Less Accurate |
| Computational Cost | Higher | Lower |

**Conclusion**

The **Canny edge detector** is more suitable because it detects **fine surface cracks** accurately while reducing false edges caused by noise.

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**C. Post-Processing Technique**

After edge detection, some crack segments may appear broken.

**Suitable Technique**

**Morphological Closing**

**Closing = Dilation + Erosion**

**Why?**

- Connects broken crack segments.
- Fills small gaps between crack pixels.
- Produces continuous crack structures.
- Improves the final crack detection result.

Additional techniques such as **connected component analysis** can also be used to remove isolated noise and retain only meaningful crack regions.

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**Final Answers (Summary)**

**A. Preprocessing Pipeline**

1. Image Acquisition
2. Illumination Correction (Histogram Equalization/CLAHE)
3. Gaussian Noise Reduction
4. Contrast Enhancement (CLAHE/Contrast Stretching)

5. Image Normalization
6. Edge Detection

**B. Edge Detection**

- **Recommended: Canny Edge Detector**
- **Comparison:** Canny provides better noise suppression, thinner edges, and more accurate localization than Sobel, making it ideal for detecting fine cracks.

**C. Post-Processing**

- Apply **Morphological Closing (Dilation followed by Erosion)** to connect broken crack segments and fill small gaps, resulting in continuous and more accurate crack detection.

**2. Human Motion Tracking in a Security Camera Video**

**Scenario:**

A security camera monitors people moving in a hallway. The system estimates motion between consecutive video frames for tracking and activity analysis.

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**A. Lucas-Kanade Optical Flow Method**

The **Lucas-Kanade optical flow** method estimates the motion of objects by tracking the movement of pixels between two consecutive video frames.

**Working Principle:**

1. Capture two consecutive video frames.
2. Detect feature points (such as corners) on the moving person.
3. Assume that the brightness of a pixel remains constant between frames.
4. Consider a small neighborhood (window) around each feature point.
5. Compute the motion vector (optical flow) by solving the displacement of pixels within the window.

6. The collection of motion vectors indicates the direction and speed of the person's movement.

**Advantages:**

- Fast and computationally efficient.
- Works well for small and smooth movements.
- Suitable for real-time human tracking.

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### B. Aperture Effect and Its Impact on Motion Estimation

#### Aperture Effect

The **aperture effect** occurs when motion is observed through a small window, making it difficult to determine the true direction of movement.

#### Effect on Uniform Regions

In areas such as:

- Plain walls
- Smooth floors
- Clothing with little texture

there are very few distinctive features. As a result:

- Motion can only be estimated along the visible edge.
- The true motion direction cannot be accurately determined.
- Optical flow vectors become inaccurate or ambiguous.

#### Result

The tracking system may:

- Produce incorrect motion estimates.
- Lose track of the moving person.
- Generate unstable optical flow vectors.

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### C. Effect of Sudden Changes in Speed or Direction

If a person suddenly changes speed or direction:

- Optical flow vectors change rapidly in both **magnitude** and **direction**.
- Vectors become longer when speed increases.

- Vectors become shorter when speed decreases.
- The orientation of vectors changes according to the new direction of movement.
- Some vectors may become inaccurate because the assumption of small motion is violated.

**Method to Improve Tracking Accuracy**

A suitable method is the **Pyramidal Lucas-Kanade Optical Flow**.

**Why?**

- Uses multiple image resolutions (image pyramid).
- Can track larger and faster movements.
- Handles sudden motion changes more accurately.
- Improves robustness in real-time tracking.

Other methods such as the **Kalman Filter** can also be used to predict a person's next position and smooth the tracking results.

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**Final Answers (Summary)**

**A.**

The **Lucas-Kanade optical flow** method tracks feature points between consecutive frames by assuming constant pixel brightness and computing motion vectors within small neighborhoods. It is efficient and suitable for real-time human motion tracking.

**B.**

The **aperture effect** occurs when there is insufficient texture (e.g., plain walls, smooth floors, or clothing), making the true motion direction difficult to estimate. This leads to ambiguous or incorrect optical flow vectors and reduced tracking accuracy.

**C.**

When a person suddenly changes speed or direction:

- Optical flow vectors change in **length (magnitude)** and **orientation (direction)**.
- Motion estimation may become inaccurate.

To improve tracking accuracy, use **Pyramidal Lucas-Kanade Optical Flow**, which handles larger displacements and rapid motion changes more effectively. A **Kalman Filter** can also be used to predict motion and provide smoother, more reliable tracking.

### 3. Vehicle Tracking at a Traffic Junction Using Video Frames

**Scenario:**

A smart traffic monitoring system uses CCTV footage to track vehicles at a busy road junction. Vehicles move at different speeds, stop at signals, turn left or right, and may be affected by shadows, reflections, and partial occlusion.

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### A. Optical Flow for Vehicle Motion Estimation

**Optical flow** estimates the movement of pixels between consecutive video frames.

**Working:**

1. Capture two consecutive video frames.
2. Detect feature points on each vehicle (e.g., corners or edges).
3. Compare the positions of these feature points in the next frame.
4. Compute the **optical flow vectors**, which represent pixel displacement.

**Motion Information:**

- **Direction:** The direction of the optical flow vectors indicates whether the vehicle is moving straight, turning left, or turning right.
- **Speed:** The length (magnitude) of the vectors indicates the vehicle's speed.
    - Longer vectors → Faster movement.
    - Shorter vectors → Slower movement or stopping.

Thus, optical flow helps estimate both the **direction** and **speed** of moving vehicles.

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### B. Two Challenges Affecting Motion Estimation

**1. Partial Occlusion**

- Occurs when one vehicle overlaps another.

- Some feature points become hidden.
- Optical flow may lose track of the vehicle or assign incorrect motion vectors.
- This reduces tracking accuracy.

**2. Shadows**

- Moving shadows create intensity changes that appear similar to moving objects.
- Optical flow may incorrectly detect shadow movement as vehicle motion.
- This produces false motion vectors and inaccurate tracking.

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### C. Improvement Method for Reliable Tracking

A suitable method is **Kalman Filter-based tracking combined with Optical Flow**.

**Why?**

- Predicts the next position of each vehicle even during temporary occlusion.
- Handles sudden changes in speed and direction more smoothly.
- Reduces the effect of noisy or missing motion measurements.
- Improves tracking continuity when vehicles overlap or turn.

For even better performance, optical flow can also be combined with **object detection (e.g., YOLO)** to re-identify vehicles after occlusion.

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### Final Answers (Summary)

**A.**

Optical flow estimates vehicle motion by tracking pixel movement between consecutive frames. The **direction** of the flow vectors indicates the direction of travel, while the **magnitude** (length) of the vectors indicates the vehicle's speed.

**B.**

1. **Partial Occlusion:** Overlapping vehicles hide feature points, causing incorrect or missing optical flow vectors and reduced tracking accuracy.
2. **Shadows:** Moving shadows create false motion, leading to incorrect optical flow vectors and inaccurate vehicle tracking.

**C.**

Use a **Kalman Filter with Optical Flow** to predict vehicle positions during occlusion and sudden direction changes, resulting in more reliable and continuous tracking.

Combining it with object detection further improves tracking accuracy.